

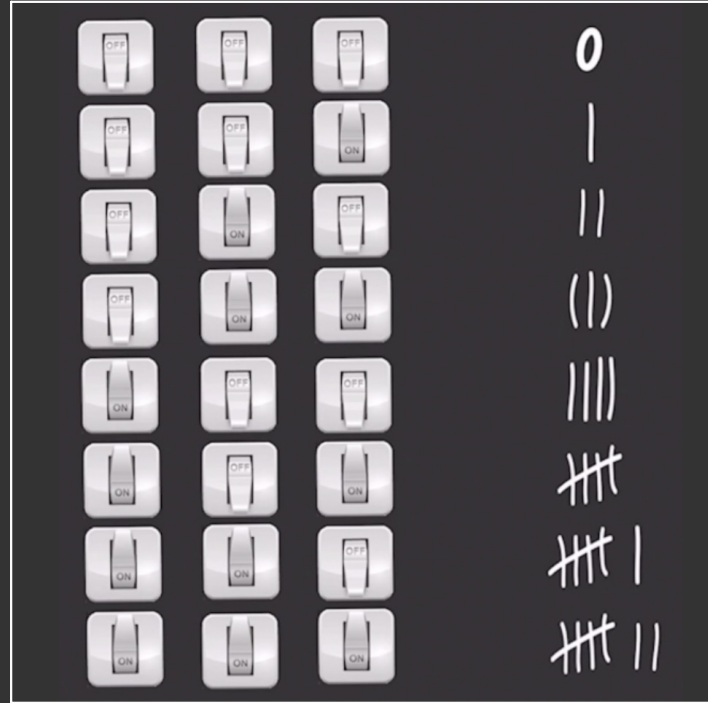
alternative
bases

CSCI
134

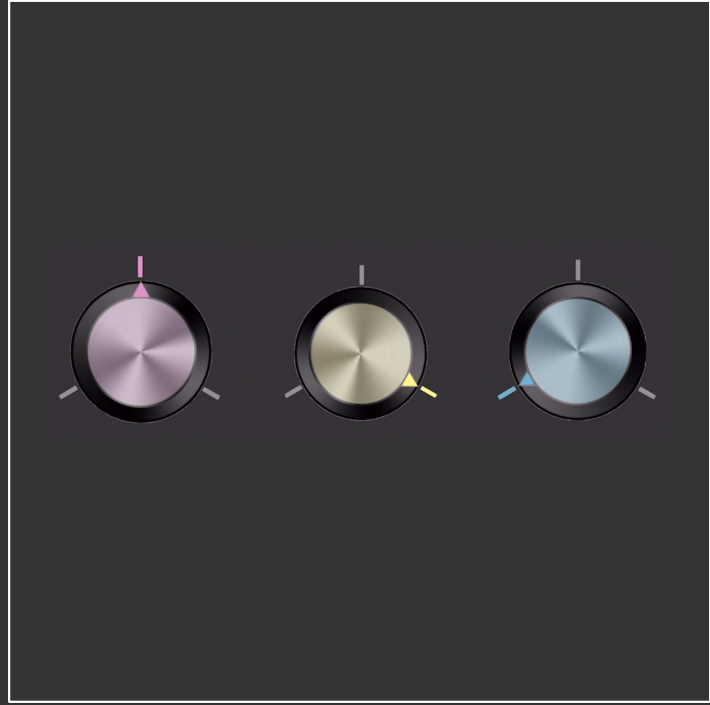
we can represent

$$2^3 = 8$$

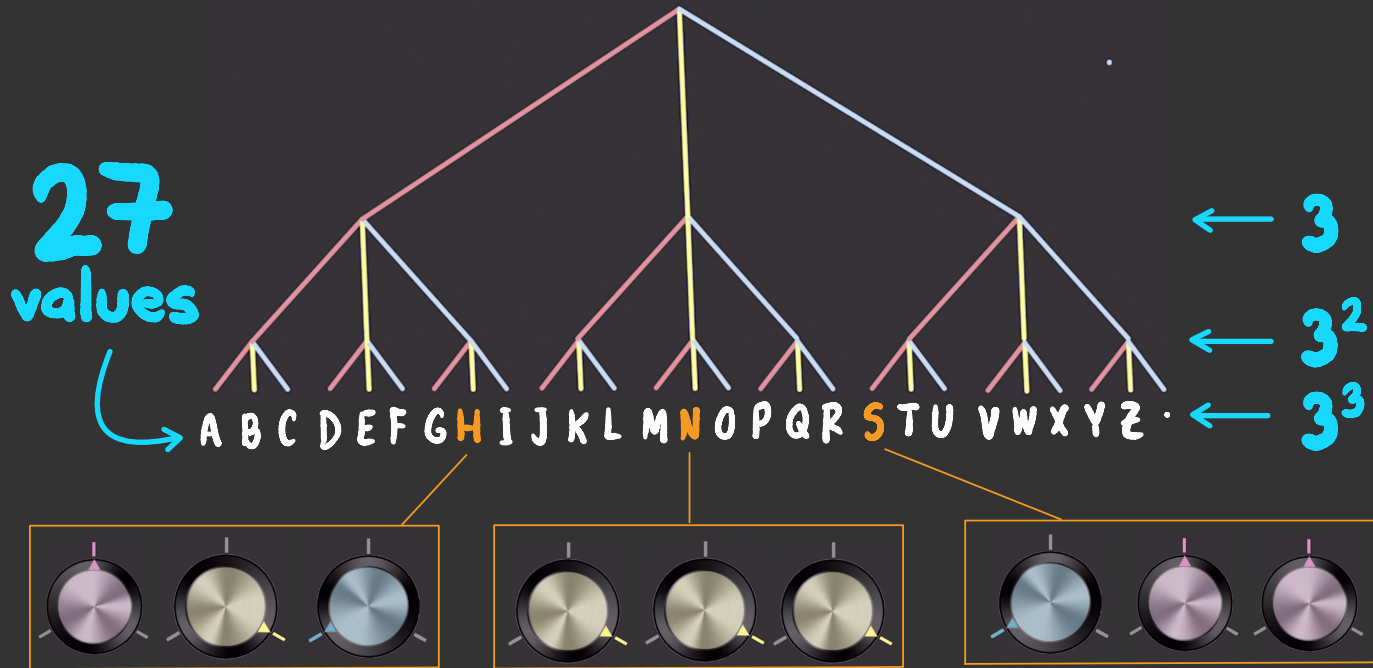
values with 3
on/off switches



how many
values can we
represent with
3 three-way dials?



how many values can we represent
with **3** three-way dials?

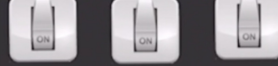


recall:

a standard representation

of "off" is zero

and "on" is one

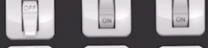
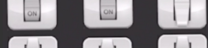
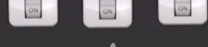
quantity	switch representation	digital representation
0		○○○
		○○
		○ ○
()		○
		○○
###		○
###		○
###		
###()	n/a	n/a

we can extend this to
three-way dials
 by adding a third digit



quantity	dial representation	digital representation
0		00
1		01
11		02
(1)		10
111		11
		12
1		20
11		21
(1)		22

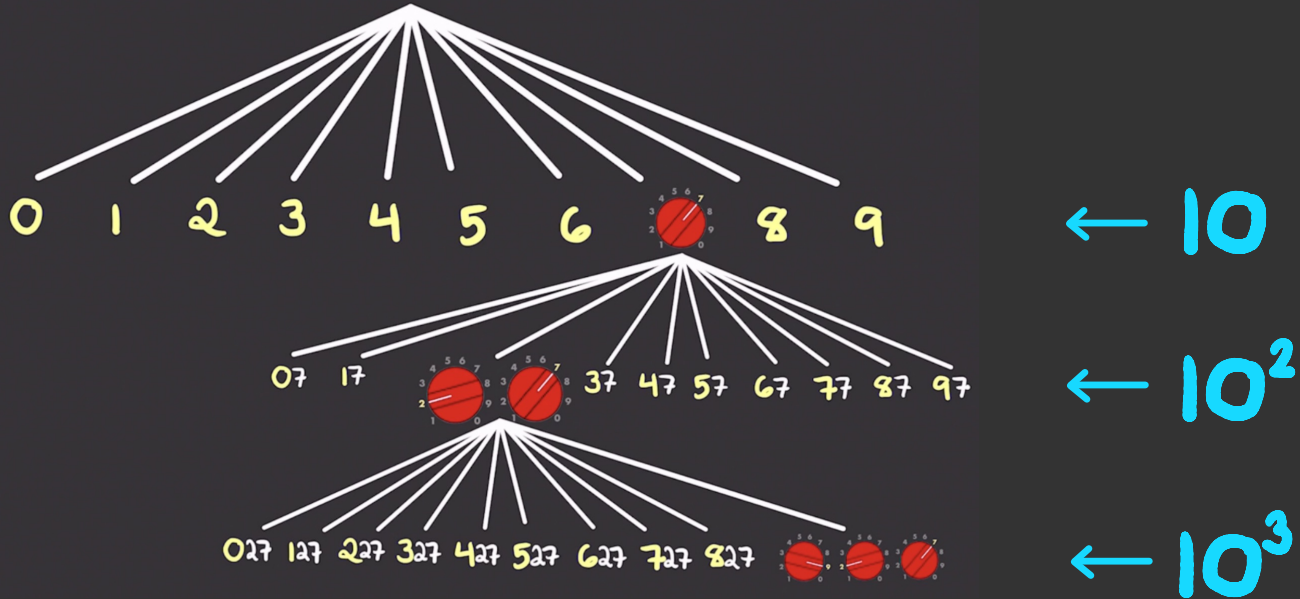
a digital representation using **k** digits
is called a base **k** representation

quantity	switch representation	base 2 representation	dial representation	base 3 representation
0				
1				
11				
(1)				
111				
1111				
11111				
111111				
1111111	n/a	n/a		

how many values can we represent
with **3** ten-way dials?



we can represent 1000 values
with 3 ten-way dials



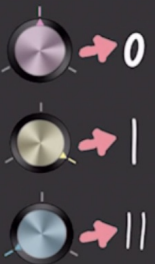
how to build a base k number system

- ▶ start with k symbols
(called digits)
- ▶ assign a unique quantity
(from 0 to $k-1$)
to each digit

base 2



base 3



base 10



how to interpret the value of a base **k** number

base **10**

0	→	0
1	→	
2	→	
3	→	()
4	→	
5	→	
6	→	
7	→	
8	→	
9	→	

2 5



how to interpret the value of a base **k** number

base **10**

0	→	0
1	→	
2	→	
3	→	()
4	→	
5	→	
6	→	
7	→	
8	→	
9	→	

10^1

tens
column

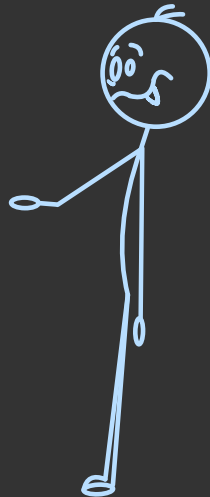
2

10^0

ones
column

5

?



how to interpret the value of a base **k** number

base **10**

0 → 0
1 → |
2 → ||
3 → (|)
4 → ||||
5 → #####
6 → #####|
7 → #####||
8 → #####(|)
9 → #####|||

10^1

tens
column

2

x #####

#####

10^0

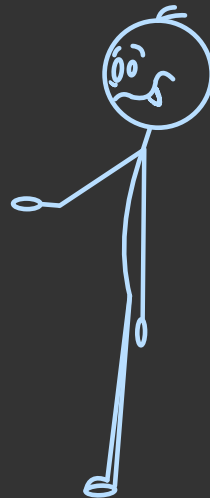
ones
column

5

x |

#####

?



how to interpret the value of a base **k** number

base **10**

0 → 0
1 → |
2 → ||
3 → (|)
4 → ||||
5 → #####
6 → #####|
7 → #####||
8 → #####(|)
9 → #####|||

10¹

tens
column

2

x #####

10⁰

ones
column

5

x |

#####

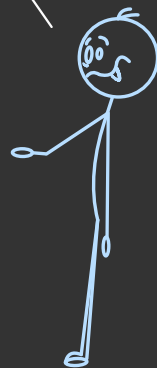
+

###

=

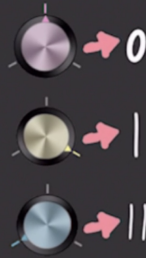
#####

25 = #####



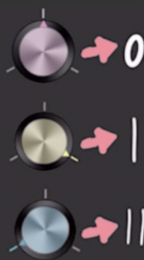
how to interpret the value of a base **k** number

base **3**



how to interpret the value of a base k number

base 3



3^2
nines
column



3^1
threes
column

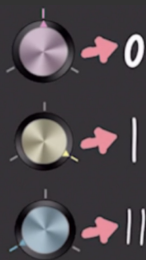


3^0
ones
column



how to interpret the value of a base **k** number

base **3**



3^2

nines
column



x ~~||||~~ ||||

~~||||~~ ||||
~~||||~~ (1)

3^1

threes
column



x (1)

|||| 1

3^0

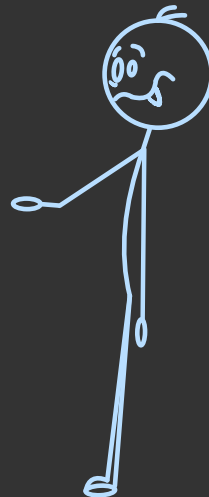
ones
column



x |

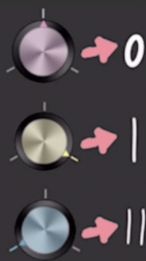
|

?



how to interpret the value of a base **k** number

base **3**



3^2
nines
column

3^1
threes
column

3^0
ones
column



x

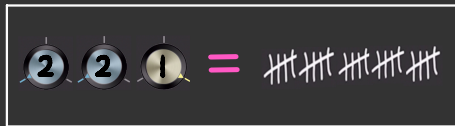
x

x

+

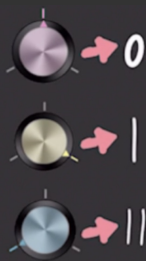
+

=



how to interpret the value of a base **k** number

base **3**



3^2
nines
column

3^1
threes
column

3^0
ones
column



x

x

x

+

+

=



how to interpret the value of a base **k** number

base **2**

0 → 0
1 → 1

1

1

0

0

1



how to interpret the value of a base **k** number

base **2**

0 → 0
1 → 1

2^4

sixteens
column

1

2^3

eights
column

1

2^2

fours
column

0

2^1

twos
column

0

2^0

ones
column

1



how to interpret the value of a base **k** number

base **2**

0 → 0
1 → 1

2^4

sixteens
column

1

x 

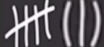




2^3

eights
column

1

x  (1)

 (1)

2^2

fours
column

0

x 

2^1

twos
column

0

x 

2^0

ones
column

1

x 



how to interpret the value of a base **k** number

base **2**

0 → 0
1 → 1

2^4

sixteens
column

1

x 

2^3

eights
column

1

x 

2^2

fours
column

0

x 

2^1

twos
column

0

x 

2^0

ones
column

1

x 



+



+

+

+



=



base **2** numbers
are also called
binary numbers



understanding
base k numbers
is vital to fully
appreciating
cs department
t-shirts

